

4 Array comparative genome hybridisation (aCGH) is a technique involving the use of a microarray to analyse a genome or sections of a genome.

(a) Outline the steps required to prepare the genome of an individual so that the genome is ready for analysis using a microarray chip.

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(b) DiGeorge syndrome is a dominant inherited disease in humans. DiGeorge syndrome is caused by deletion of a large number of nucleotides from chromosome 22.

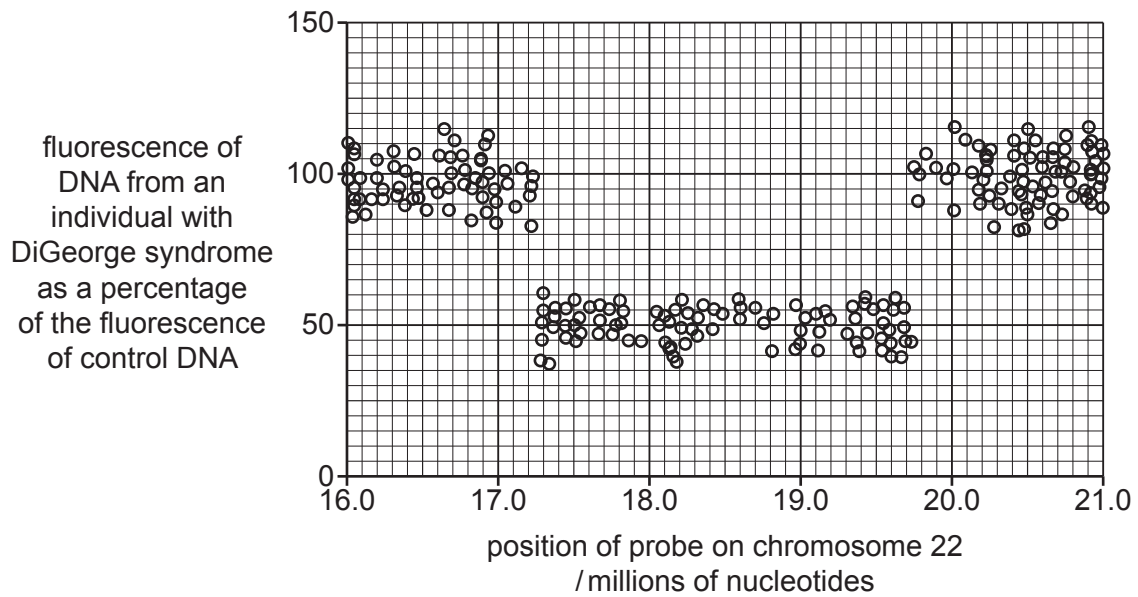
The number of nucleotides deleted varies between individuals in a range from 800 000 to 3 100 000. The largest deletions can cause the removal of up to 46 protein-coding genes from the chromosome.

Fig. 4.1 shows the results of aCGH using a microarray specific for the section of chromosome 22 within which the DiGeorge syndrome deletion occurs. The microarray analysed DNA from two individuals:

- one with DiGeorge syndrome
- one who did not have DiGeorge syndrome (control DNA for comparison).

In the aCGH results shown in Fig. 4.1:

- Each small circle represents the results from a single probe on the microarray.
- The x-axis shows the position of each probe on chromosome 22. The position is shown as distance along the chromosome in millions of nucleotides.
- A result close to 100% fluorescence on the y-axis means that the DNA from the individual with DiGeorge syndrome fluoresces at the same intensity as the control DNA for that probe.
- A result close to 50% fluorescence on the y-axis means that the DNA from the individual with DiGeorge syndrome fluoresces half as much as the control DNA for that probe.



**Fig. 4.1**

- (i) With reference to Fig. 4.1, estimate the number of nucleotides deleted from the affected chromosome 22 in the individual with DiGeorge syndrome.

Give your answer to the nearest 100 000 nucleotides.

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- (ii) Explain how the microarray technique works to give the results shown in Fig. 4.1.

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(iii) Suggest why the phenotypes of two individuals with DiGeorge syndrome can be different.

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[Total: 10]

**Question 5 starts on page 16.**